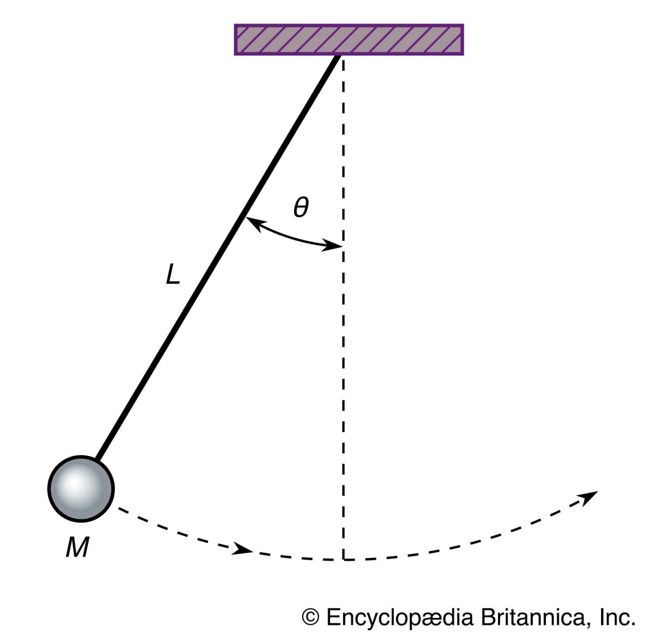
Learning Dynamical Systems From Data

Axle Zhao

APMA 2070

Prof. George Karniadakis

Background - Pendulum



Hamiltonian equations of motion:

$$\frac{dp_\theta}{dt} = -mgl\sin\theta$$

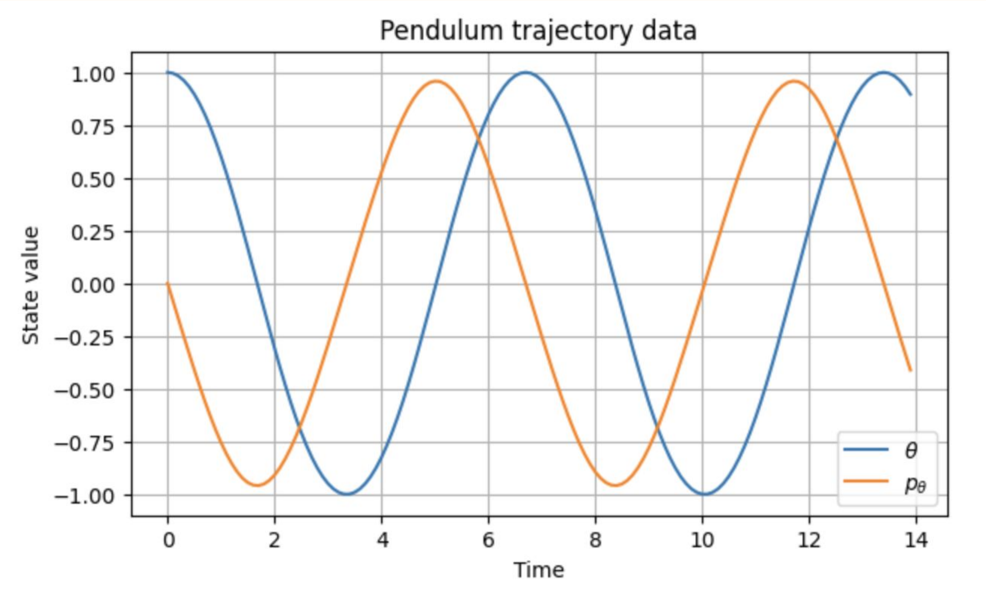
$$\frac{d\theta}{dt} = \frac{p_\theta}{ml^2}$$

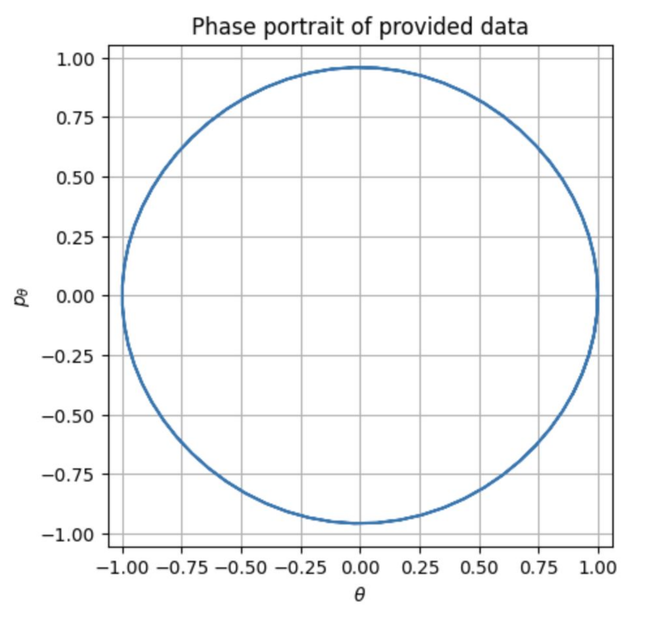
g: gravitational constant
L: pendulum length
θ: angular displacement
pθ: angular momentum

[1]https://www.britannica.com/technology/pendulum

Background - Dataset

- Non dimensionalized system with m = l = g = 1
- 140 total time steps: train.txt (n=40), test.txt (n=100). Initial conditions [pθ(0), θ(0)] = [0, 1].
- Trajectory data generated using the symplectic Störmer–Verlet integrator (h = 0.1)
- Chosen because it preserves Hamiltonian structure and long-time energy stability.





Physics Sanity Checks

Hamiltonian:

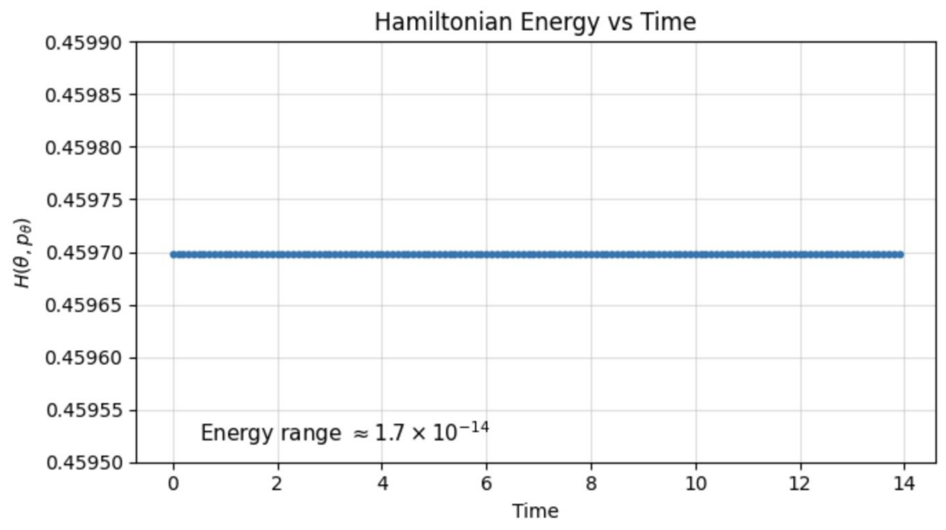
$$H(\theta, p_\theta) = \frac{p_\theta^2}{2ml^2} + mgl(1 - \cos\theta)$$

1st term: kinetic energy
2nd term: gravitational potential energy

m = l = g = 1:

$$H(\theta, p_\theta) = \frac{1}{2}p_\theta^2 + 1 - \cos\theta$$

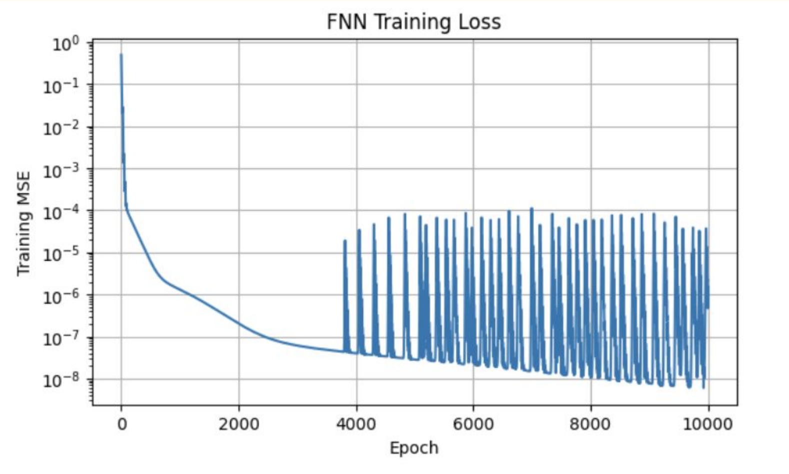
For a frictionless pendulum, the total Hamiltonian energy should remain constant over time.

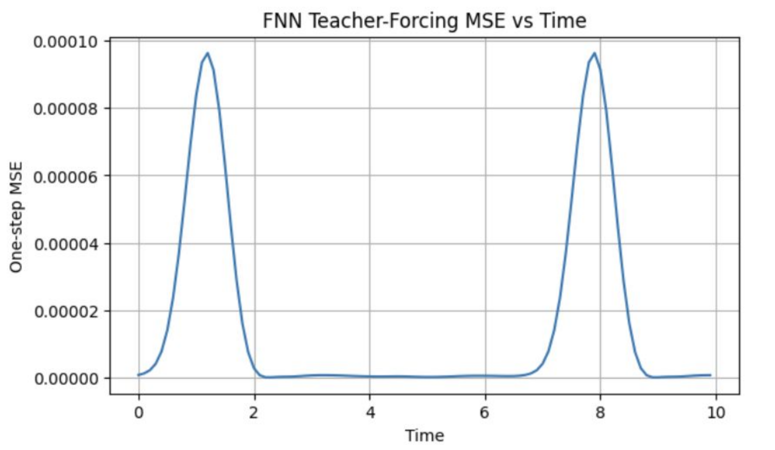


FNN

$$F_\phi(x_i) \approx x_{i+1}, \qquad x_i = \begin{bmatrix} p_{\theta,i} \\ \theta_i \end{bmatrix}$$

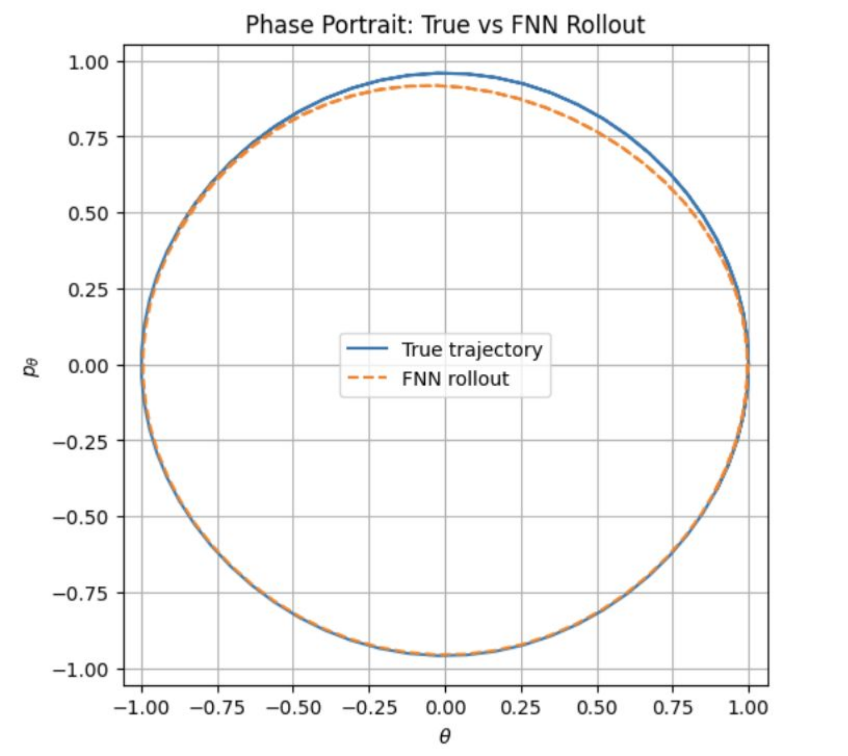
- Feedforward neural network trained on one-step transitions
- Loss: MSE
- Optimizer: Adam, learning rate = 10^{-3}

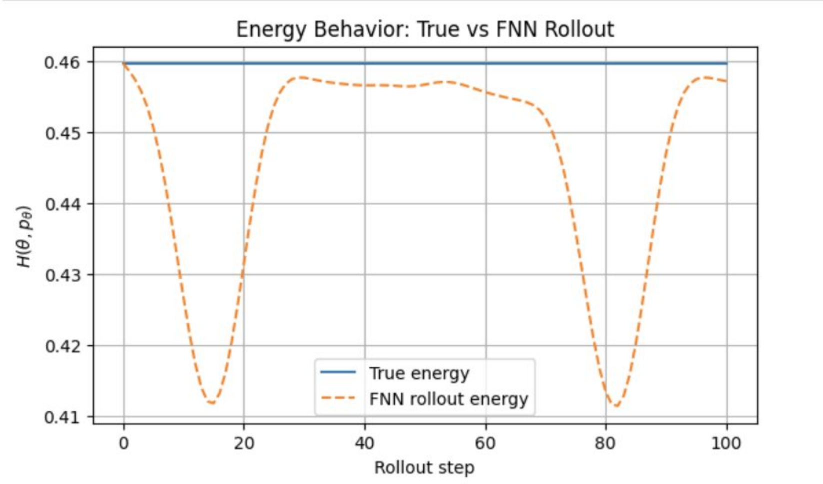




- Mean teacher-forcing MSE: 1.66×10^{-5}
- FNN accurately learns local one-step dynamics.

FNN





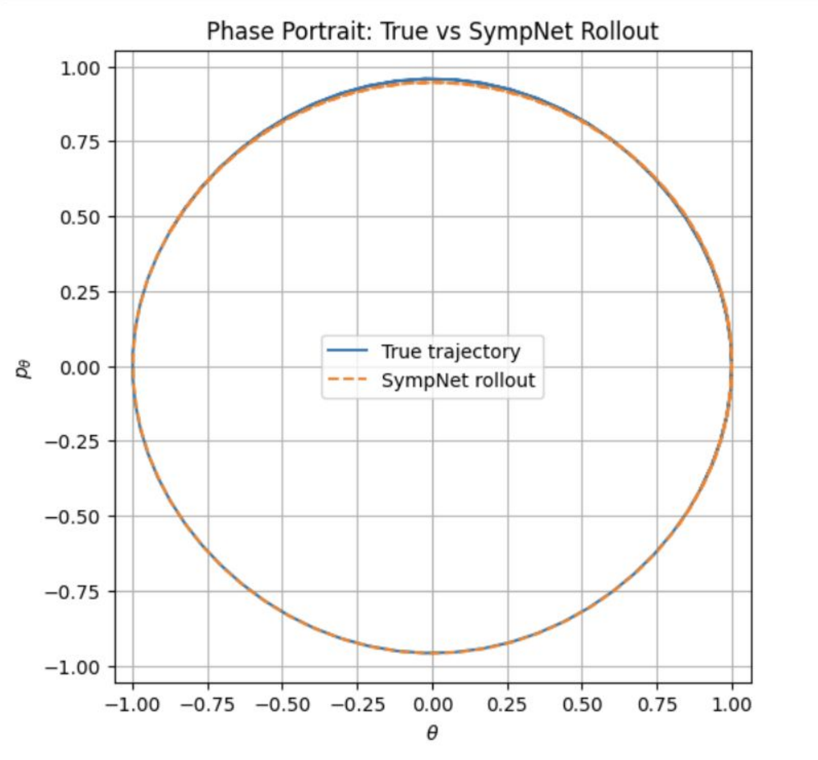
- One-step errors accumulate during rollout
- Rollout drifts away from the true Hamiltonian orbit
- FNN does not explicitly preserve Hamiltonian structure

Energy variation: 4.82×10^{-2} (vs 10^{-14} for true dynamics)

SympNet

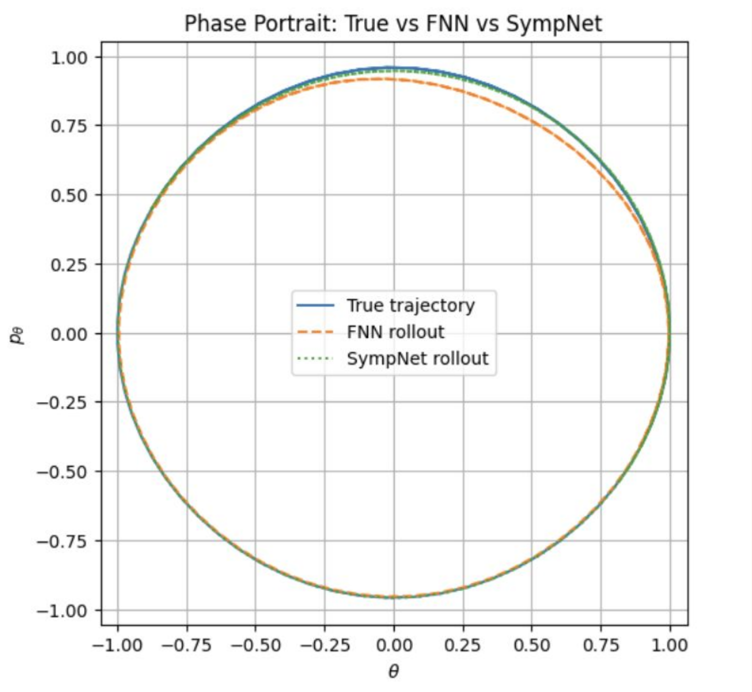
$$x_i \mapsto x_{i+1}, \qquad x_i = (p_\theta, \theta)$$

$$D\Phi(x)^T J D\Phi(x) = J$$



- SympNet enforces symplectic structure in phase space
- Built from alternating symplectic shear maps
- Goal: improve long-term rollout stability

SympNet



- SympNet preserves orbit geometry better
- Energy drift reduced:
  ΔH:
  FNN $\approx 4.8\times10^{-2}$
  SympNet $\approx 1.5\times10^{-2}$
- Rollout MSE remains larger than FNN
- Better physics consistency does not guarantee lower pointwise error

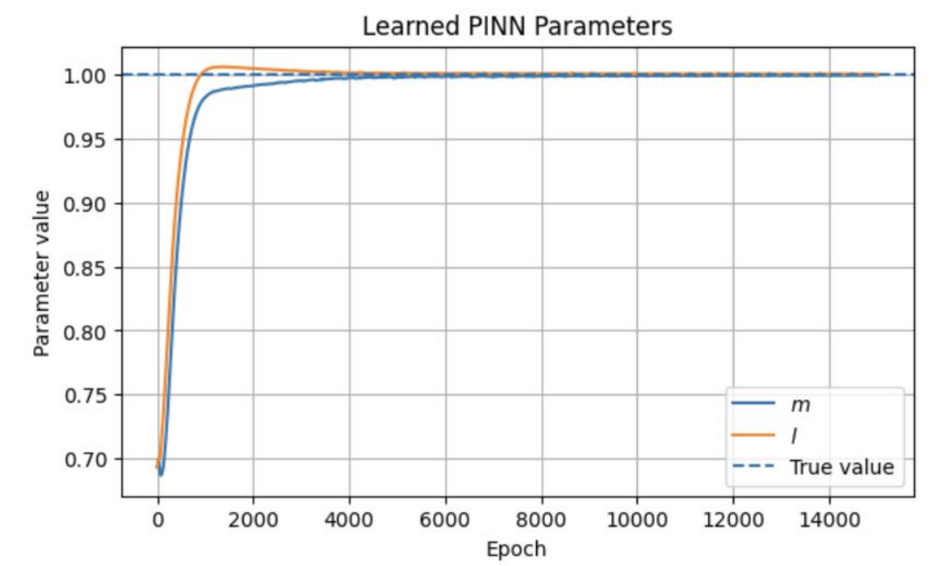
Structure preservation improves physical consistency, but not necessarily rollout MSE.

PINN

$$H(\theta, p_\theta) = \frac{1}{2ml^2}p_\theta^2 + mgl(1 - \cos\theta)$$

$$\dot{\theta} = \frac{\partial H}{\partial p_\theta}, \qquad \dot{p}_\theta = -\frac{\partial H}{\partial \theta}$$

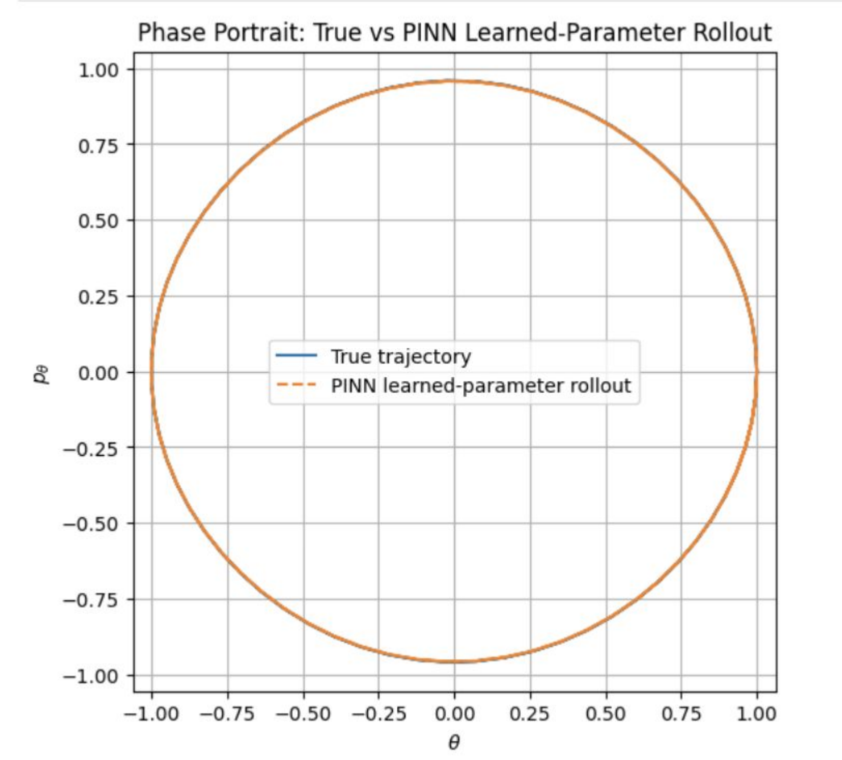
$$\mathcal{L} = \mathcal{L}_{data} + \lambda \mathcal{L}_{physics}$$

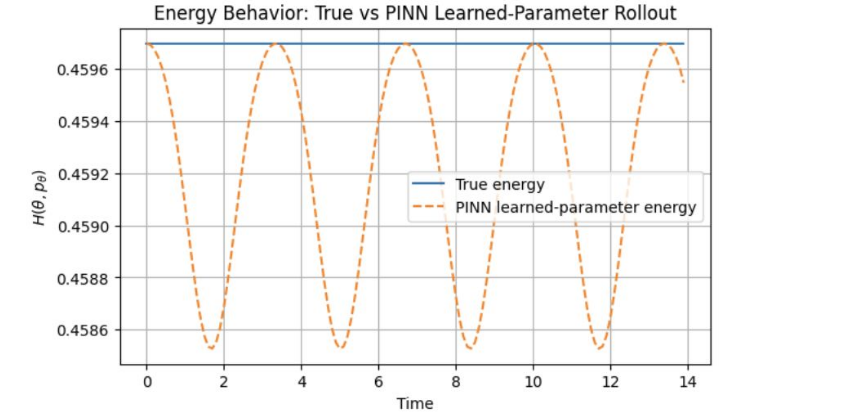


- PINN learns physical parameters m and l from trajectory data
- Loss combines data fitting + Hamiltonian physics residual
- Learned: m ≈ 0.9994, l ≈ 1.0003
- Parameters converge close to true values

Trajectory data are sufficient to recover the true pendulum parameters.

PINN

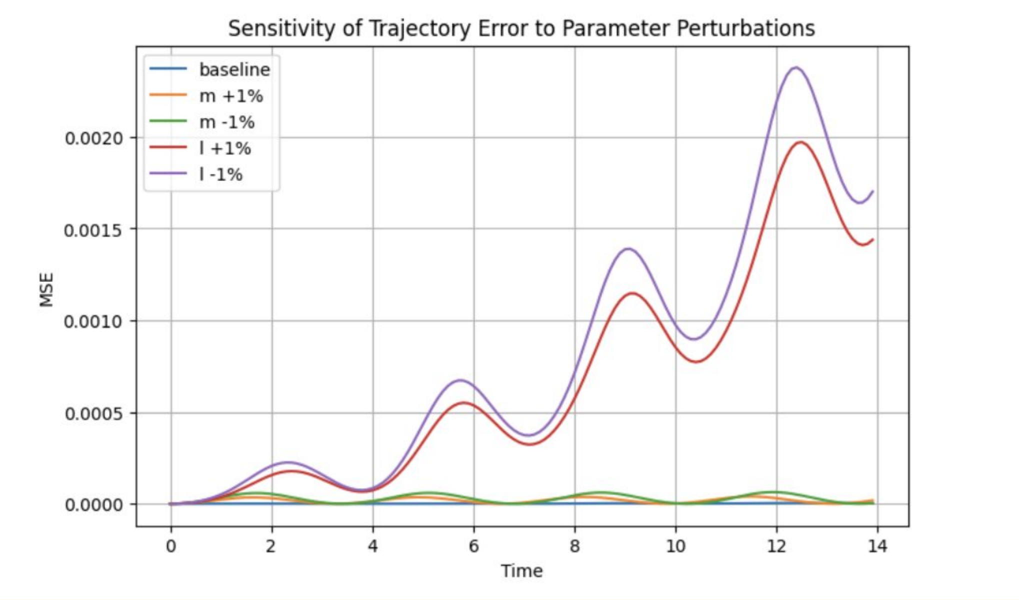


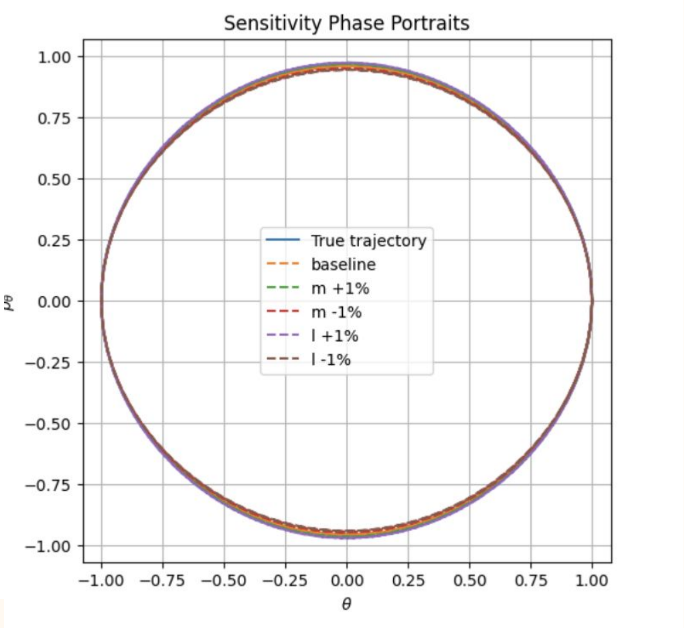


- PINN rollout closely matches the true orbit
- Mean rollout MSE: 1.45 × 10^-6
- Energy drift remains very small: ΔH ≈ 1.17 × 10^-3

PINN recovers both accurate trajectories and physically consistent dynamics.

PINN





- ±1% parameter perturbations cause growing trajectory error
- Length perturbations affect rollout more strongly than mass perturbations
- Small parameter errors accumulate over long horizons

Small parameter uncertainty can produce large long-time trajectory deviations.

Comparative Analysis: FNN vs SympNet vs PINN

Property	FNN	SympNet	PINN
Inductive bias	Low	Medium	High
Physics structure	None	Symplectic geometry	Hamiltonian dynamics
Data efficiency	Lower	Medium	Higher
Long-time energy behavior	Weak	Good	Best
Rollout MSE (this experiment)	Best	Worse than FNN	Best overall
Robustness to distribution shift	Lower	Medium–High	Highest

Increasing physical inductive bias improves consistency and robustness, but stronger structure does not always minimize rollout error.

Thank you!